



Greece — Monthly Intelligence Report

v4.2

Edition 006 — The Thessaloniki Industrial Grid Corridor

Where the energy transition is outrunning the grid. edition · August 2026 · SSI v4.2
(refreshed June 2026)

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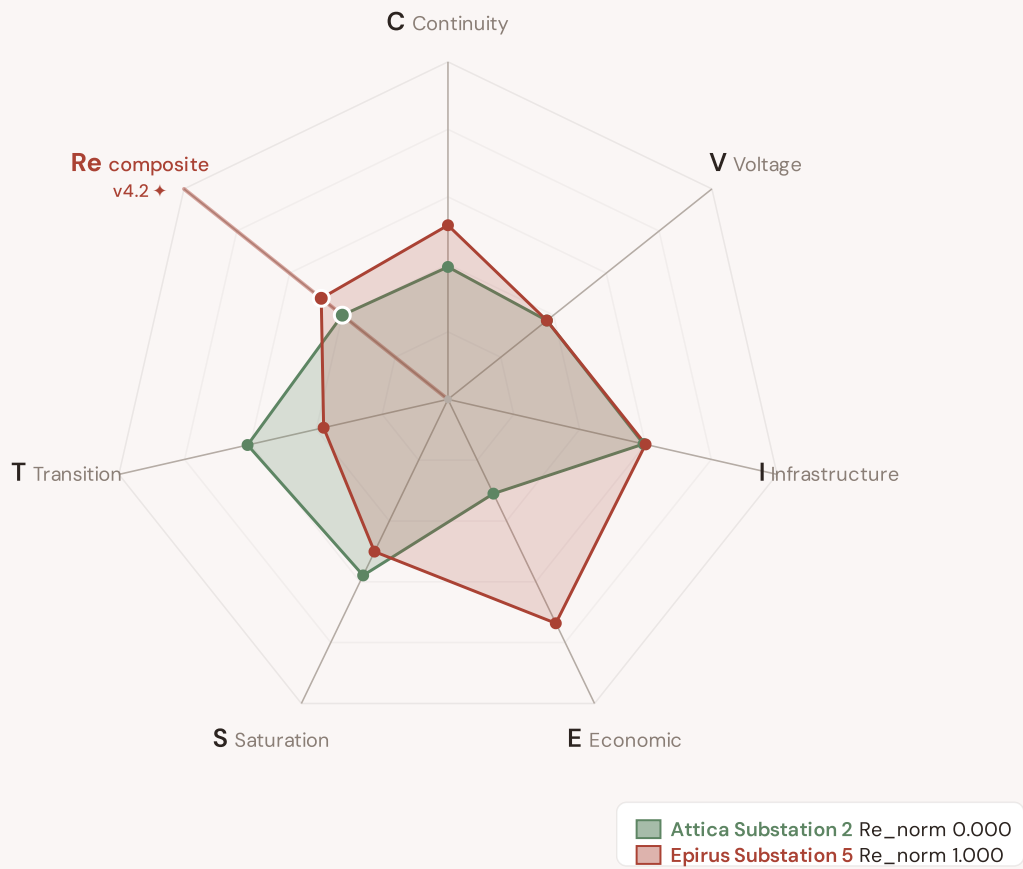
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SECTION A

Executive Summary

Radar 1 — v4.2: now 7-axis with the new Re composite

v4.0.2 Radar 1 had 6 axes (C/V/I/E/S/T). **v4.2 adds Re — the bounded resilience composite** capturing the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in one normalised scalar ($Re_norm \in [0, 1]$ via $\text{clip}((Re_raw - 0.920) / (1.787 - 0.920), 0, 1)$). Two worked examples overlaid: **Attica Substation 2** (low-exposure reference, Re_norm 0.000) · **Epirus Substation 5** (high-exposure reference, Re_norm 1.000). The Re axis is colour-highlighted in cayenne to emphasise the v4.2 architectural addition.



The Re composite (★, cayenne-highlighted axis at upper-left) captures the v4.2 resilience modifier family in one bounded scalar — formula $Re_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$, bounds $[0.920, 1.787]$. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are algorithmically selected (min/max Re_norm) per the v4.2 methodology. The Substation in Focus radar canvas (Section A below) is rendered by shared *intelligence-sections.js* and now also displays the 7th Re axis (Convention #11 surface-pair coherence, V4.2-disc-24 closure 12 Jun 2026).

SUBSTATIONS SCORED

719

65 EHV · 422 HV · 232
distribution-tier

GRID LINES MAPPED

1,775

~12,400 km coverage

MC SIMULATIONS

42.9 M

10,000 per substation

PUBLIC DATA SOURCES

34

101 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 42.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 244,701 output data points with full uncertainty quantification across 719 substations and 1,775 grid lines spanning ~12,400 km. v4.2 adds the Re composite (7th axis) and six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just) on top of the v4.0.2 foundation.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 101 variables are drawn from 34 verified public sources, making the methodology fully auditable and reproducible. Initially covering Greece's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

South Aegean Substation 530

South Aegean, Andros, Thira, Kea – Kythnos, Milos, Mykonos, Naxos, Paros, Syros, Tinos · 150 kV

0.792

Critical

0.694

87th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE

Re composite v4.2 7th axis

Re_raw 1.2737 · Re_norm 0.069

LOW EXPOSURE

COMPONENTS

C

Continuity: 1.000 / 0.30 (333%) ▲

I

Infrastructure: 0.451 / 0.25 (181%)

S

Saturation: 1.000 / 0.20 (500%) ▲

V

Voltage: 1.000 / 0.10 (1000%) ▲

E

Economic: 0.379 / 0.10 (379%)

T

Transition: 0.883 / 0.05 (1766%)

MODIFIERS

R3

Consequence: 0.955 ↓ dampens

R4

Graph Criticality: 0.965 ↓ dampens

R6a

Restoration: 1.080 ↑ amplifies

R6b

Seismic: 1.100 ↑ amplifies

R7

Digital Readiness: 1.012 ↑ amplifies

This month's spotlight — automatically selected for September 2026 — is **South Aegean Substation 530** in South Aegean, Andros, Thira, Kea – Kythnos, Milos, Mykonos, Naxos, Paros, Syros, Tinos. With an R_final of 0.792 (Critical band, 87th fleet percentile), its risk profile is dominated by the T (Transition) component at 1766% of its weight ceiling, followed by V (Voltage) at 1000%. Modifiers collectively shift R_base by 0.1%, reflecting the substation's specific geographic, socio-economic, and network context.

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that infrastructure digitalisation planning requires.

R7 v2 CRA-anchored transition (Q3 2026 → Q1 2027 dual-write): the R7 signal is entering a dual-write window where the legacy R7_cyber (DESI + ACN scalar proxy) co-exists with R7_cyber_v2 (CRA Article 14 + NIS2 Article 21 register-anchored composite, Path C+D). Consumers may select either; substation audit trail carries _r7_cyber_v2_source when v2 populated.

CYBER HIGH EXPOSURE

248

34.5% of fleet

BLIND SPOTS

177

High/Critical/Extreme + R7 < median

AVG R7 MODIFIER

1.0082

Range [0.99, 1.05]

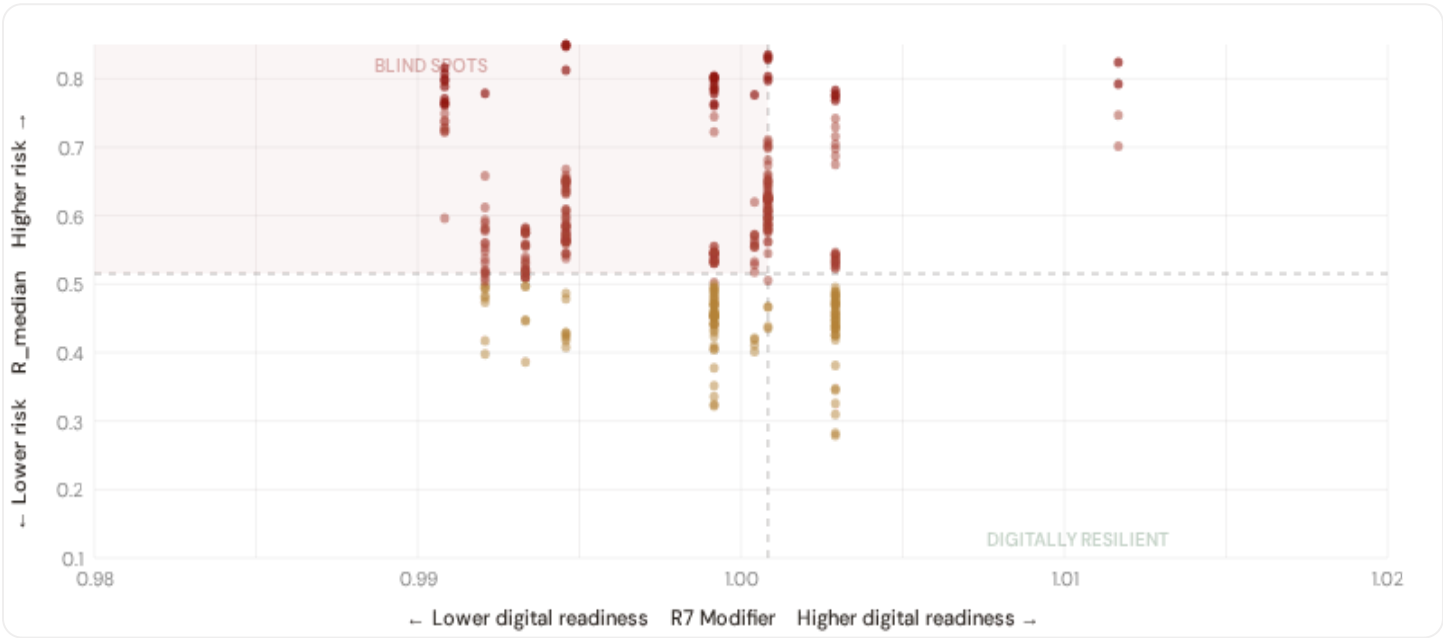
HIGH-CONSEQUENCE SUBS

140

19.5% · R3 ≥ 1.08

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



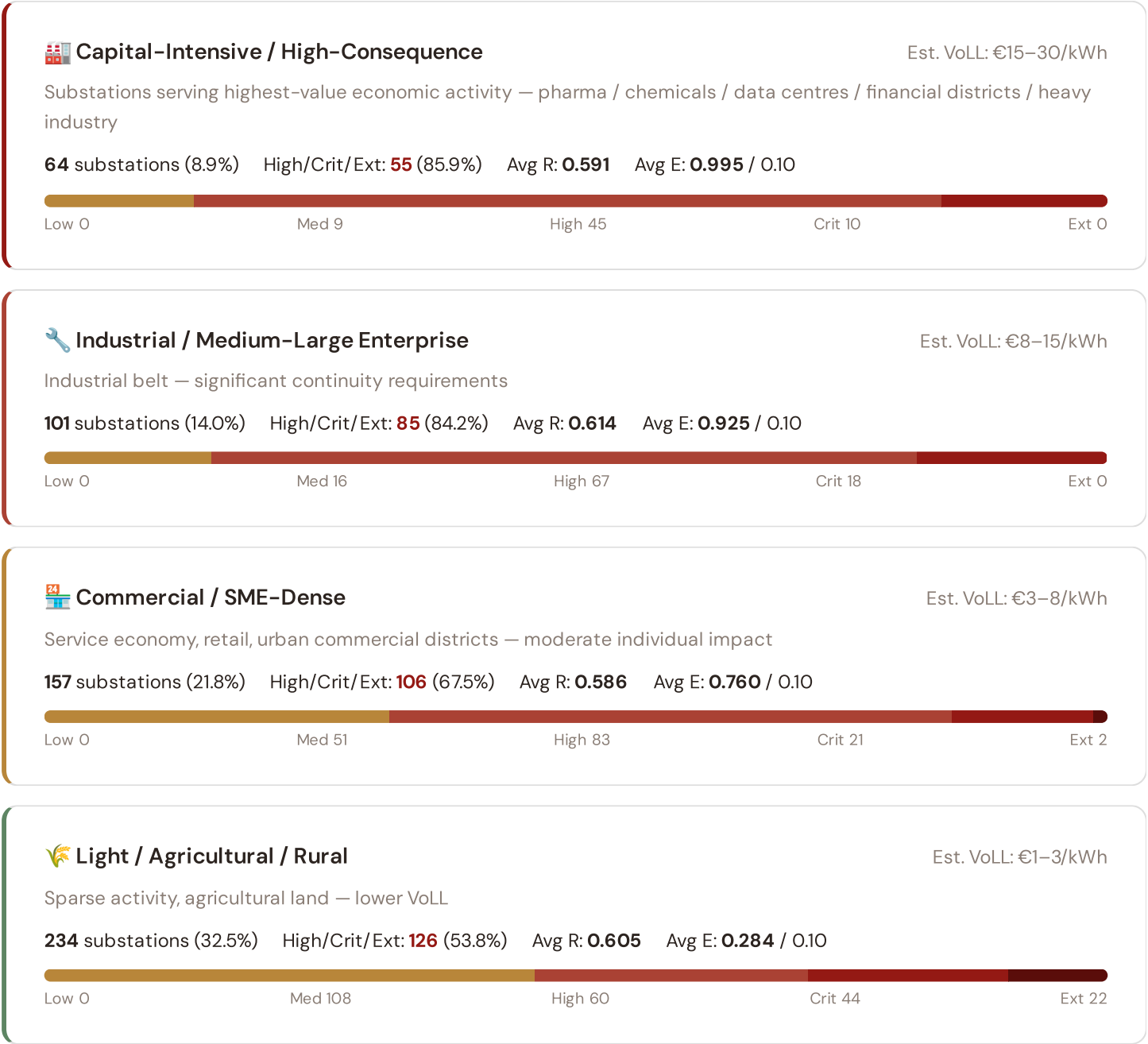
● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **177 blind-spot substations** — scoring High, Critical, or Extreme on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0008$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Grevena, Kozani (20), Aitolokarnania (14), Larisa (14)**. Across the full fleet, 248 substations (34.5%) carry a HIGH cyber-exposure classification, while 131 (18.2%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

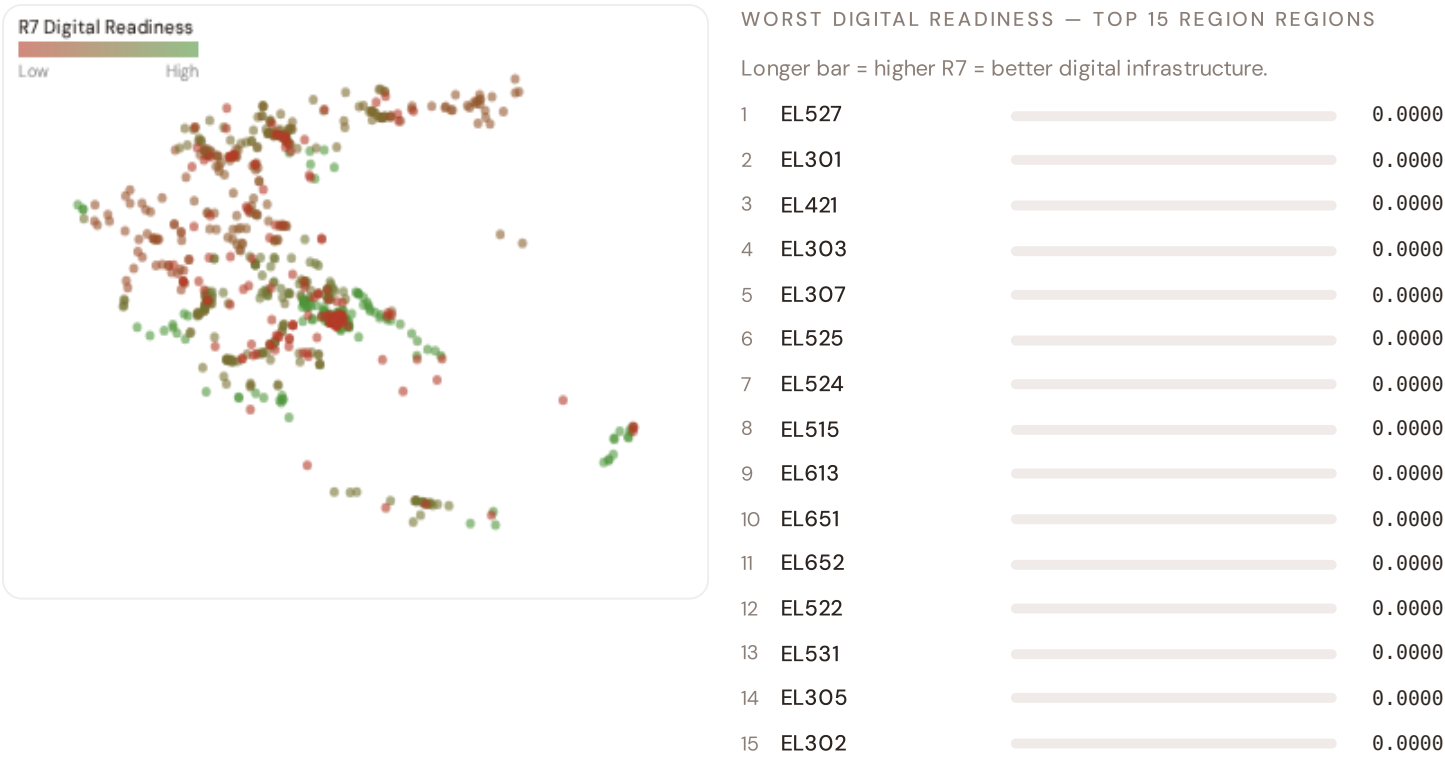


Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's Economic component (CVIESTR-E) segments substations by the economic value they serve. **55 substations** combine capital-intensive economic fabric (top-decile Economic component ($E \geq 0.990$, P90+)) with High, Critical, or Extreme risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Rodopi (14), Aitolokarnania (13), Evros (10)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 24.6%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Province Digital Readiness Ranking

Provinces ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_{median} to identify provinces combining digital exposure with high grid stress.



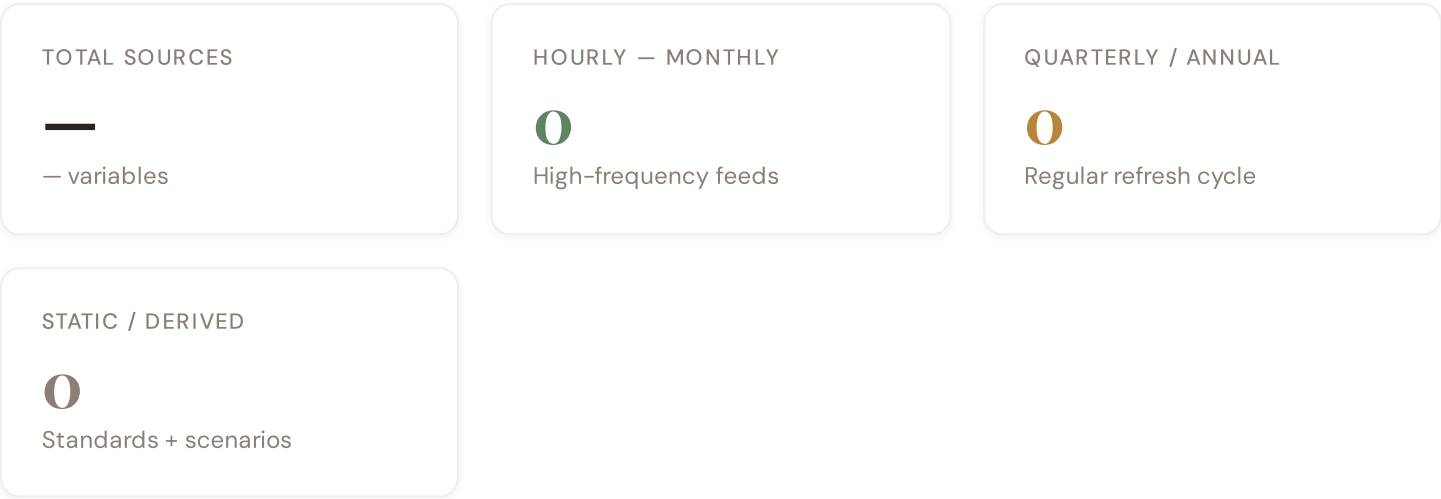
region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **EL527** has the lowest average R7 (0.0000), while **Attica** leads at 1.0392 — a spread of 1.0392 across the R7 range. **8 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (September 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0082, median = 1.0008; 248 substations classified HIGH cyber-exposure; 177 blind-spot substations (High/Critical/Extreme risk + below-median R7); 42 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

Data Refresh & Changelog

The SSI v4.0.2 draws on — **verified public data sources** feeding — variables across — components and — modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.



Data Source Registry — September 2026

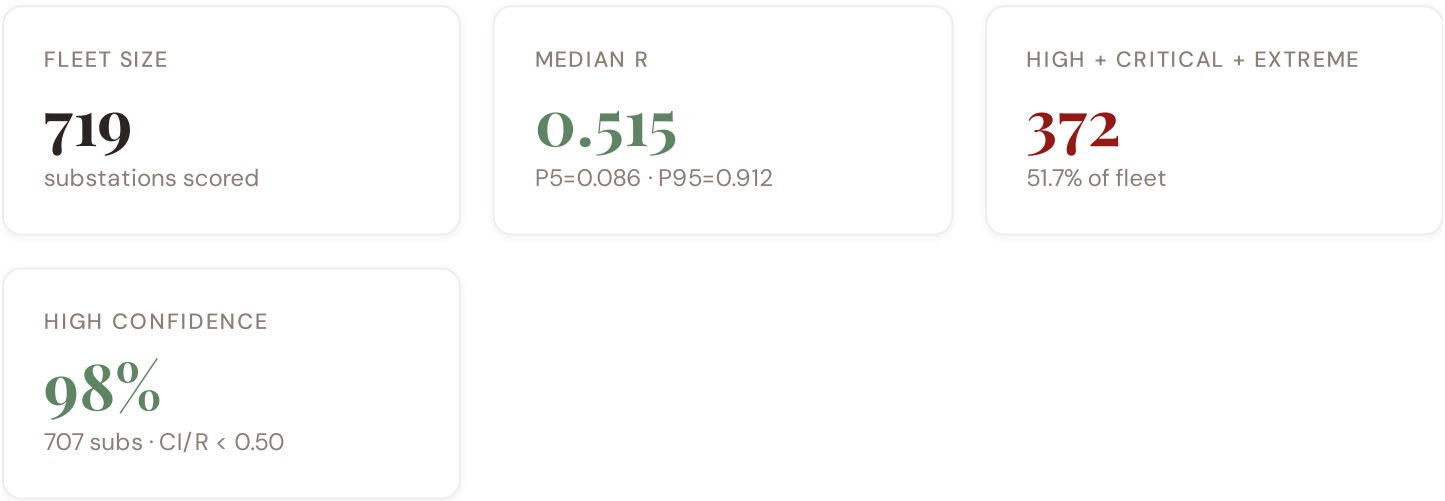
OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
HMEPATHS	Hellenic Meteorological Service	DAILY	~5 km	3 vars
DEDDIE	DEDDIE/HEDNO (Διανομή Ηλεκτρισμού)	MONTHLY	Prefecture	5 vars
PDVSA	Public Power Corporation Data	MONTHLY	Prefecture	2 vars
RAE	RAE (Ρυθμιστική Αρχή Ενέργειας)	QUARTERLY	Prefecture	3 vars
IPTO-EX	IPTO Extended Topology	QUARTERLY	Prefecture	3 vars
HEPI	HEPI (Hellenic Energy Policy Index)	ANNUAL	National	3 vars
EUROSTAT	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
ETVA	ETVA (Ελληνικό Ταμείο Αναπτύξεως)	ANNUAL	Prefecture	2 vars
MINENV	Ministry of Environment	ANNUAL	Prefecture	2 vars
HMEPD	Hellenic Forest Management	ANNUAL	Prefecture	1 var
INFOSTAT	Infrastructure Statistics	ANNUAL	Prefecture	1 var
APIVITA	Industry Association Data	ANNUAL	Prefecture	1 var
COPERNICUS	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
IEEE-1	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars
EAK2003	EAK 2003 Seismic Code	STATIC	Prefecture	2 vars
GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
COPDEM	Copernicus DEM / Landcover	STATIC	30 m	2 vars
DIMOVSKI	Dimovski et al. (2025)	STATIC	Municipal	3 vars
ISLAND-ISO	Island Isolation Factor	STATIC	Prefecture	1 var
COG-GR	COG Prefecture Registry	STATIC	Prefecture	1 var

ADMIE	ADMIE/ IPTO (Ανεξάρτητο Διαχειριστή)	REAL-TIME	Nodal	8 vars
ELSTAT	ELSTAT (Ελληνική Στατιστική Αρχή)	QUINQUENNIAL	Prefecture	8 vars
HENEX	HEnEx (Ελληνική Ηλεκτρική Ανταλλαγή)	HOURLY	National	3 vars
ENTSOE	ENTSO-E Transparency	HOURLY	Control Area	2 vars
NOA	NOA (Εθνικό Αστεροσκοπείο)	REAL-TIME	Prefecture	4 vars
MF-OM	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars
D-OM	Open-Meteo / ERA5	HOURLY	~1 km	1 var

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

C.2 Fleet Score Snapshot



No primary data sources have been updated since the v4.0.2 launch baseline. All **719 substation scores remain unchanged** this edition. The fleet median R of 0.515 (mean 0.497) reflects a distribution skewed toward the lower bands, with 21.1% in Low and 27.1% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2 → v4.2

v4.2 (June 2026) adds six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just), the Re composite (7th axis, bounds [0.920, 1.787]), four new data sources (IdroGEO PGRA, SIAB wildfire, ECMWF ERA5 Bourguoin, bulk

smart meter), six new variables, and W1–W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6) on top of the v4.0.2 baseline.

11 changes tracked across PO, PI, and P2 releases

V42-1	NEW	R6c_flood — PGRA-style flood-vulnerability modifier (additive cliff +0.00..+0.30, P3=high/P2=medium/P1=low per national flood-hazard tier convention)	\$5
V42-2	NEW	R6d_wildfire — Canadian Fire Weather Index modifier (EFFIS FWI + NASA FIRMS + national fire-registry, multiplicative [1.00, 1.20])	\$5
V42-3	NEW	R6e_winter — Bourgoiuin winter-storm modifier (ECMWF ERA5 winter-precipitation signal, multiplicative [1.00, 1.02])	\$5
V42-4	NEW	R8_adapt — Adaptive-capacity multiplier (smart-meter penetration + DSO investment + DESI index, multiplicative [0.95, 1.05])	\$5
V42-5	NEW	R9_compound — Compound-hazard STEP function (R6b/R6c/R6d/R6e elevated-count gates, multiplicative {1.00, 1.04, 1.08, 1.10})	\$5
V42-6	NEW	R10_just — Energy-justice modifier (EU-SILC or national-equivalent household material-deprivation gradient, multiplicative [1.00, 1.12])	\$5
V42-7	NEW	Re composite — bounded resilience scalar $Re_{raw} = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$; $Re_{norm} = clip((Re_{raw} - 0.920)/(1.787 - 0.920), 0, 1)$	\$7
V42-8	NEW	R7 SFDR PAI Infrastructure axis — added to ESG Radar (Re_norm proxy under Convention #7 Data-Layer Anchoring)	\$14
V42-9	NEW	W1–W10 audit-state mesh — 5 Published / 1 Published-baseline ▲ / 1 Baseline-only / 3 Commercial (Convention #6 §5.6)	\$4
V42-10	CHANGE	95 → 101 variables (+6 from v4.2 modifier inputs); 30 → 34 data sources (+4 hazard / monitoring layers)	data
V42-11	CHANGE	Substation fleet: 581 substations (per OSM canonical extract + national TSO/DSO registry cross-validation)	data

SECTION D — MONTHLY DEEP-DIVE

The Thessaloniki Industrial Grid Corridor

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** the monthly theme · **002** DER saturation hotspots · **003** Climate vulnerability under CMIP6 · **004** North–South economic impact disparity · **005** Voltage quality and industrial output · **006** Infrastructure age vs investment · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Sub-national Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Theme Context

THEME-RELEVANT SUBSTATIONS	HIGH BAND	CORRIDOR MEDIAN R
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3
3 EHV · 0 HV

0 + 3
100.0% of corridor

1.052
Fleet median: 0.515

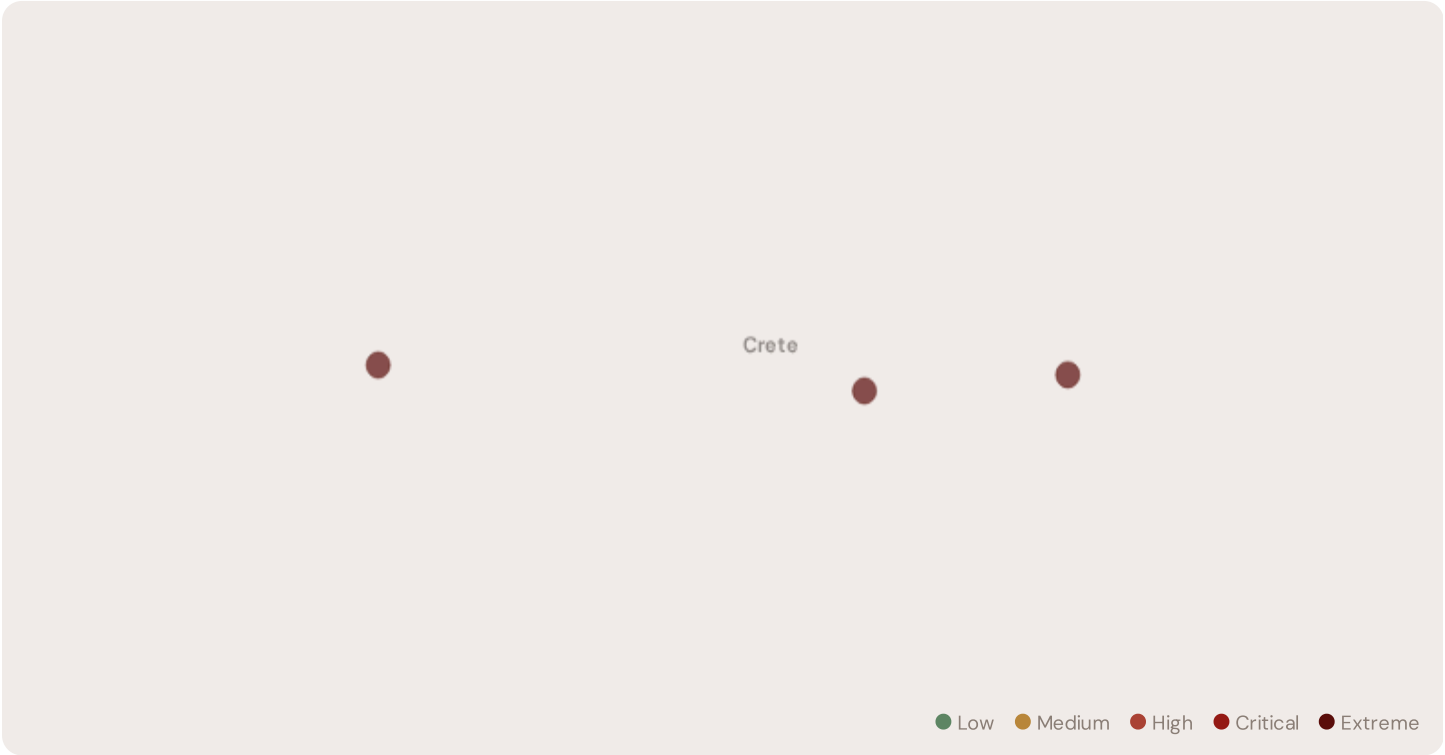
WORST PROVINCE

Crete

Avg R: 1.054 · 3 substations

The Chania hosts **3 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **0 substations (0.0%)** are classified High, with only **0** in the Low band. 100% of corridor substations score above the national fleet median of 0.515. The corridor median R of 1.052 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE v4.2	ALERT
Crete Substation 429	Crete	1.058	Extreme	0.875	1.000	0.754	0.611	1.000	0.841	S, V	
Crete Substation 464	Crete	1.052	Extreme	1.000	1.000	0.754	0.611	1.000	0.841	S, V	

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE v4.2	ALERT
Crete Substation 468	Crete	1.052	Extreme	0.875	1.000	0.754	0.611	1.000	0.841	S, V	

... -7 additional substations — [explore full corridor in Digital Twin →](#)

D.3 Component Analysis

COMPONENT AVERAGES — CHANIA VS FLEET

C — Continuity	0.917 vs 0.360 (+154%)
I — Infrastructure	0.754 vs 0.298 (+153%)
S — Saturation	1.000 vs 0.208 (+381%)
V — Voltage	1.000 vs 0.204 (+391%)
E — Economic	0.611 vs 0.477 (+28%)
T — Transition	0.841 vs 0.227 (+271%)

MODIFIER AVERAGES — CHANIA VS FLEET

R3 Consequence	0.992	fleet 0.807 →
R4 Graph Crit.	1.000	fleet 0.767 →
R6a Restoration	1.080	fleet 0.763 ↑
R6b Seismic	1.200	fleet 1.131 ↑
R7 Digital Read.	1.002	fleet 0.780 →

The dominant risk driver across the Chania corridor is **T (Transition)**, averaging 0.841 against a fleet average of 0.227 — occupying 1681% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 1.000 vs fleet 0.204 (1000% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.992 (fleet: 0.807), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.200, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Province Breakdown

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Crete

3 subs · avg R = 1.054 · H:0 E:3 M:0

The region regions-level breakdown reveals significant intra-regional variation. **Crete** leads with a mean R of 1.054 across 3 substations, while **Crete** scores 1.054 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Chania is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

3 Chania substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Crete sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are

disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

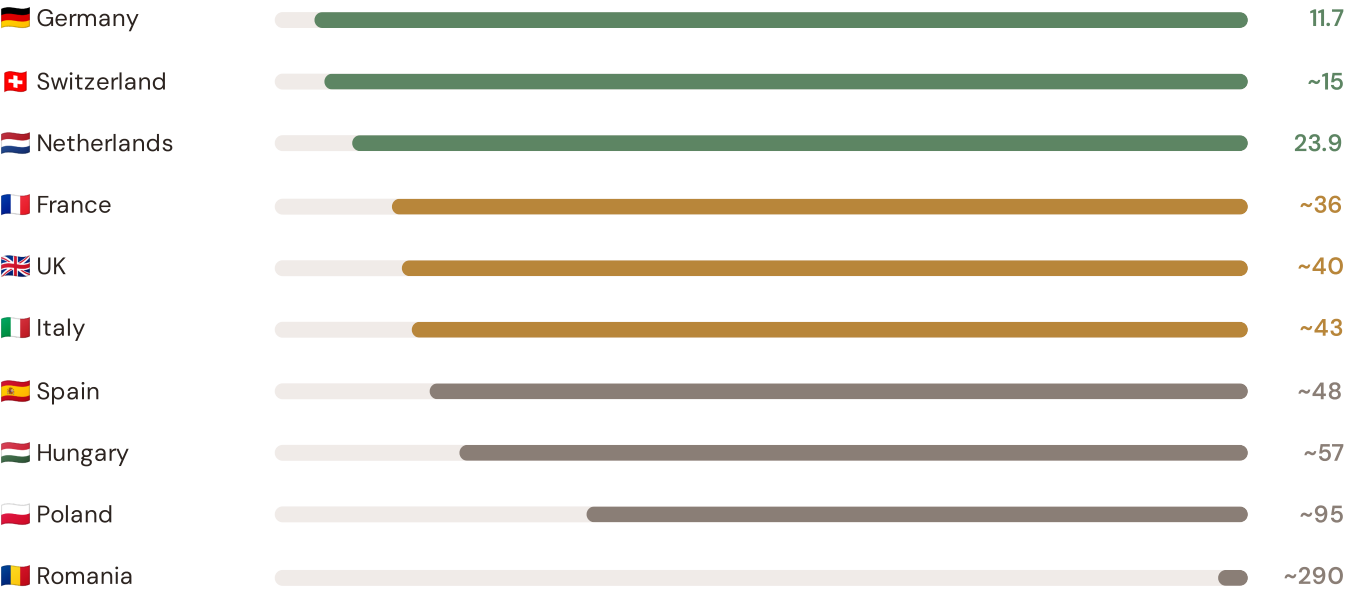
SECTION E — RECURRING MODULE

OECD Context Card

How this works: Each edition includes one OECD Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Southern Coastal corridor analysis hinges on Greece's continuity performance relative to OECD peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected OECD Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



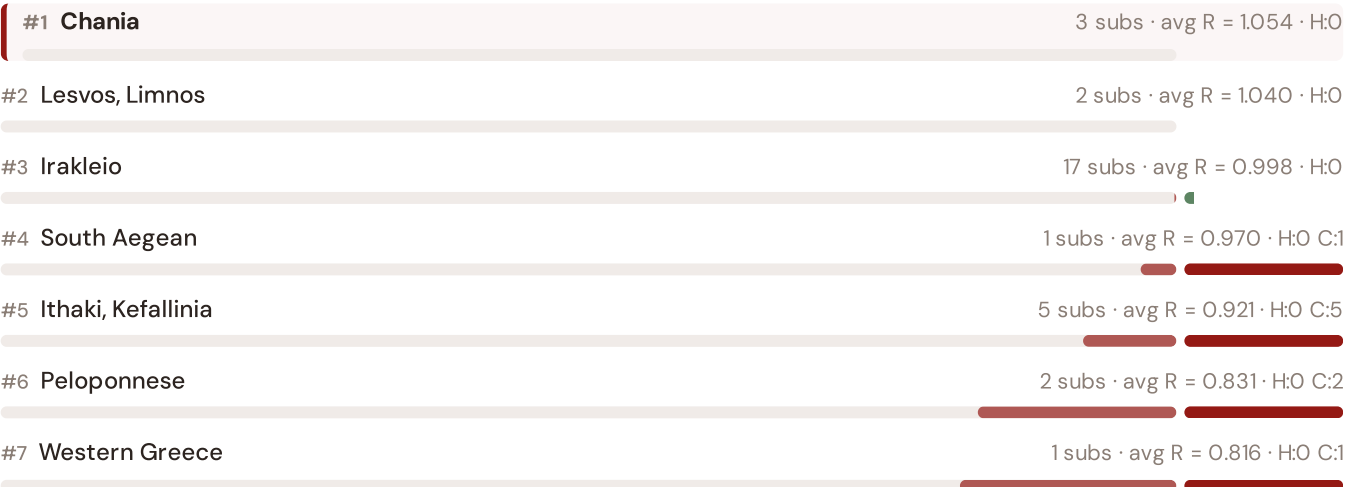
Source: CEER 7th Benchmarking Report (2022); Bundesnetzagentur (2024); ARERA (2023). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release. · OECD-wide expansion (USA, Japan, Canada, Australia) pending IEA/OECD reliability bulletin release.

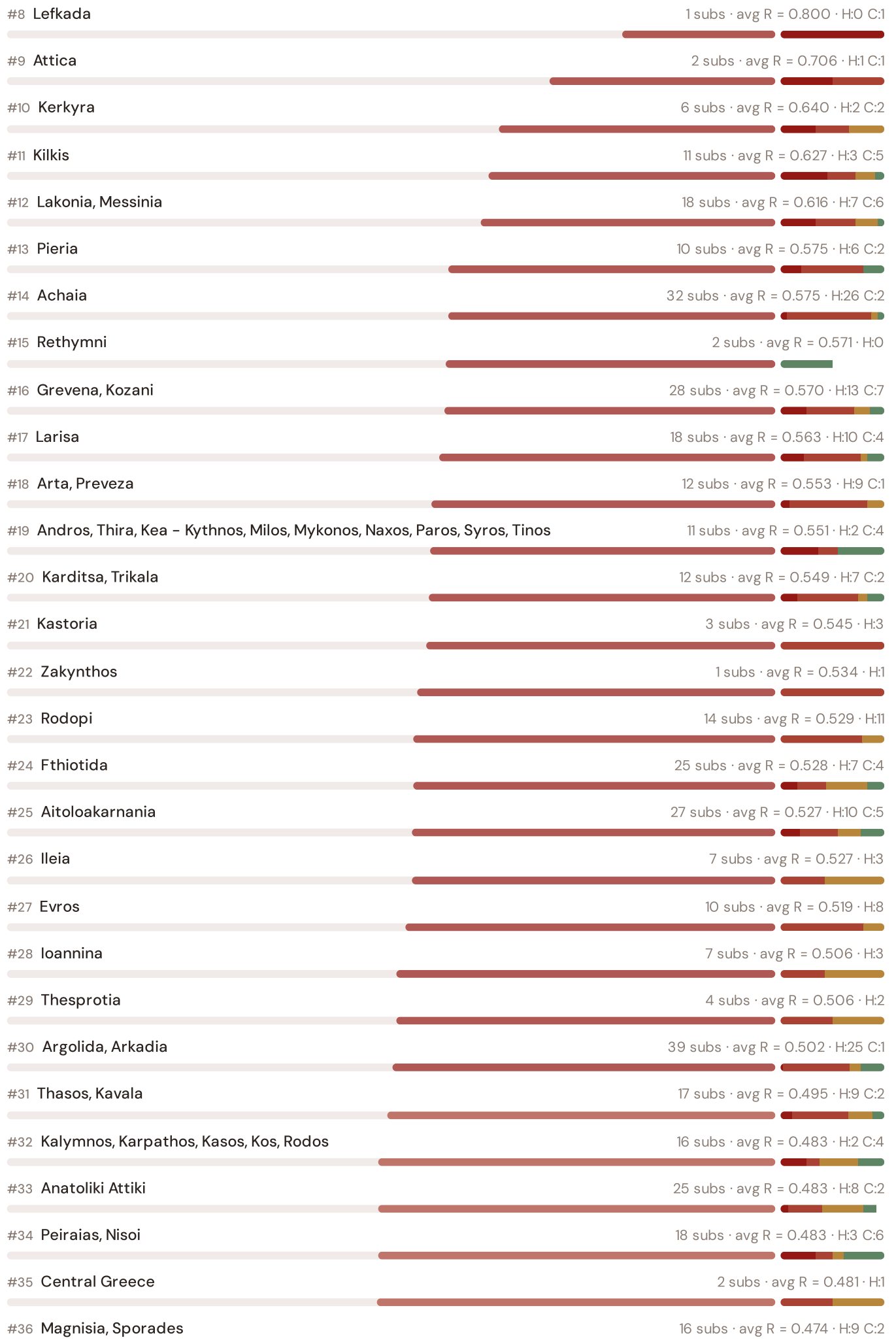
This month's key insight: The Chania corridor deep-dive shows **3 substations** (of 3) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.917 is **2.5× the fleet average** of 0.360. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regions Risk Ranking — All 55 Greek Regions

Where does this corridor sit within the national landscape? Regions sorted by mean R_{median}, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.





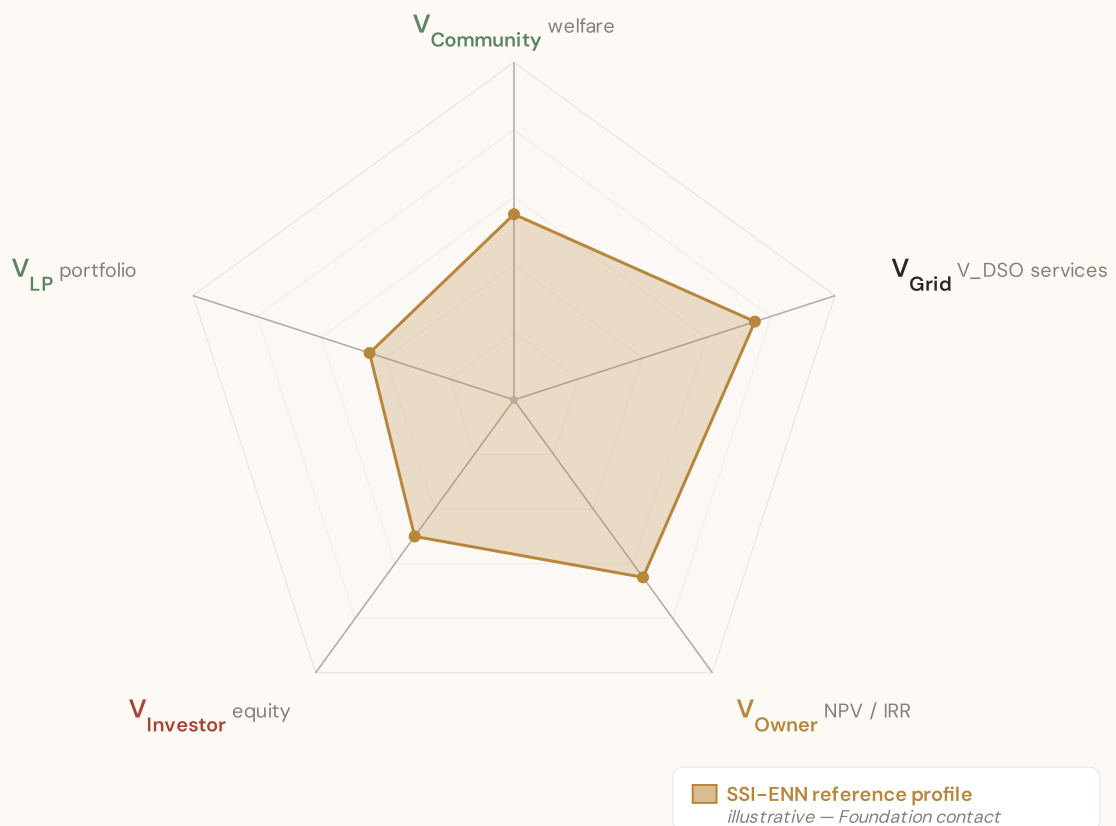


The Chania ranks **#1 of 55 region regions** by mean R_{median}, placing it among the upper tier of national risk. The three most stressed are Chania, Lesvos, Limnos, Irakleio, while the three most resilient are Kentrikos Tomeas Athinon, Voreios Tomeas Athinon, Notios Tomeas Athinon. 30 of 55 score above the fleet average of 0.497. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

From grid intelligence to BESS valuation platform. The SSI Index answers "How healthy is this substation?" The SSI Enhanced Neural Network (SSI-ENN) extends this into: "What is a BESS worth at this substation — to the DNO and to the community?" Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

SSI-ENN value-articulation radar — 5-lens stakeholder taxonomy

The SSI-ENN's commercial-tier value articulation answers "what is a BESS worth at this substation — and to whom?" through five stakeholder lenses (architecture memo §5 Decision #12): **V_{Community}** (welfare avoided outages, energy-poverty relief, employment) · **V_{Grid}** (V_DSO — avoided reinforcement, reliability, voltage, congestion, frequency, black-start, hosting capacity) · **V_{Owner}** (NPV / IRR / payback per asset class) · **V_{Investor}** (risk-adjusted equity return at the financing tier) · **V_{LP}** (limited-partner cash-on-cash + DPI / TVPI portfolio metrics). The pentagon shape below is an editorial reference profile, not per-substation commercial data — per-substation evaluation requires the SSI-ENN engines (L2-21 SupplyChainDD · L4-O5 Stakeholder Decomposition · L2-SYN Synergy). Foundation contact: ssi_index@ikenga.eu.



Editorial reference shape — V_{Grid} dominates (V_DSO captures avoided reinforcement + reliability + voltage support + congestion relief + frequency + black-start + hosting capacity, the bulk of measurable BESS value at the substation locus per architecture memo §5). V_{Community} + V_{Owner} are co-secondary. V_{Investor} + V_{LP} reflect the financing-tier waterfall above V_{Owner} NPV. Per-substation commercial decomposition lives in the SSI-ENN — mail to ssi_index@ikenga.eu for analytics.

Layer 1 — Data

SSI v4.0 component taxonomy,

Layer 2 — Valuation

DSO/TSO + Community value stacks, TCO, real options

Layer 3 — Neural Net

GBT+DNN ensemble, transfer learning, SHAP

Layer 4 — Coverage

Country prioritisation, DCI, registry, calibration

Layer 5 — Platform

Products, API, revenue model, unit economics

Gaussian Copula Monte Carlo Engine

How 10,000 simulations per substation capture the correlation structure across grid risk factors

The SSI components are not independent — a substation with poor continuity (high C) often also has poor voltage quality (high V), aged infrastructure (high I5), and is located in a rural area with high economic vulnerability (high E). The Gaussian copula captures these dependencies through a 20×20 correlation matrix estimated from the national fleet, decomposed via Cholesky factorisation to generate correlated random draws. For each substation, 10,000 Monte Carlo iterations propagate uncertainty through the full scoring formula, producing not just a point estimate (R_median) but a complete posterior distribution — P5, P50, P95, confidence interval width, and skewness. This is the computational backbone, yielding the uncertainty quantification that distinguishes the SSI from deterministic indices.

KEY CONCEPTS

→ 20×20 Gaussian copula with Cholesky decomposition

→ 10,000 iterations per substation

→ Full posterior: R_P5, R_median, R_P95, CI_width, skewness

→ Correlation captures systemic clustering of risk factors

LIVE SSI DATA CONNECTION

The SSI Index currently scores **719 substations**. Fleet mean R = 0.497, with an average confidence interval width of 0.120 from 10,000 MC iterations per substation. The dominant component is C (Continuity) at avg 0.360 / 0.30 ceiling (120% utilisation).

Coming next month:

LAYER 5

 Platform — Product Delivery Pipeline — *From raw analytical outputs to four commercial product formats*

Radar 2 — W1–W10 Substation Community Value

Wall 1 — Public W1–W10 framework. The 10-axis Radar 2 expresses per-substation community-value baseline pre-intervention: *W1 Reliability · W2 Carbon · W3 Local Economic · W4 Climate Resilience · W5 Digital Inclusion · W6 Sovereignty · W7 Local Value Retention · W8 Communications · W9 Supply Security · W10 Compute Multiplier.* The discipline floor that the anti-maladaptation governance gate protects: *"no W axis worse off after intervention."*

5/1/1/3 4-tier audit-state classification (mesh resolution per Convention #6 §5.6)

Published (5) — solid fill

W1 · W2 · W3 · W4 · W8

Peer-reviewed methodology operational at fleet scale.

Published-baseline (1) ▲ — solid fill + chevron corner marker

W9 — Supply Security

Peer-reviewed academic basis (Albert/Holme topology-only baseline) with acknowledged scope limit (Buldyrev/IEEE 493 — N-1 contingency dynamics not captured at baseline).

Baseline-only (1) — striped fill

W5 — Digital Inclusion

Foundation methodology under development; baseline value defensible at v4.2.

Commercial (3) — dashed-faded

W6 · W7 · W10

Commercial deliverables — please contact the Foundation.

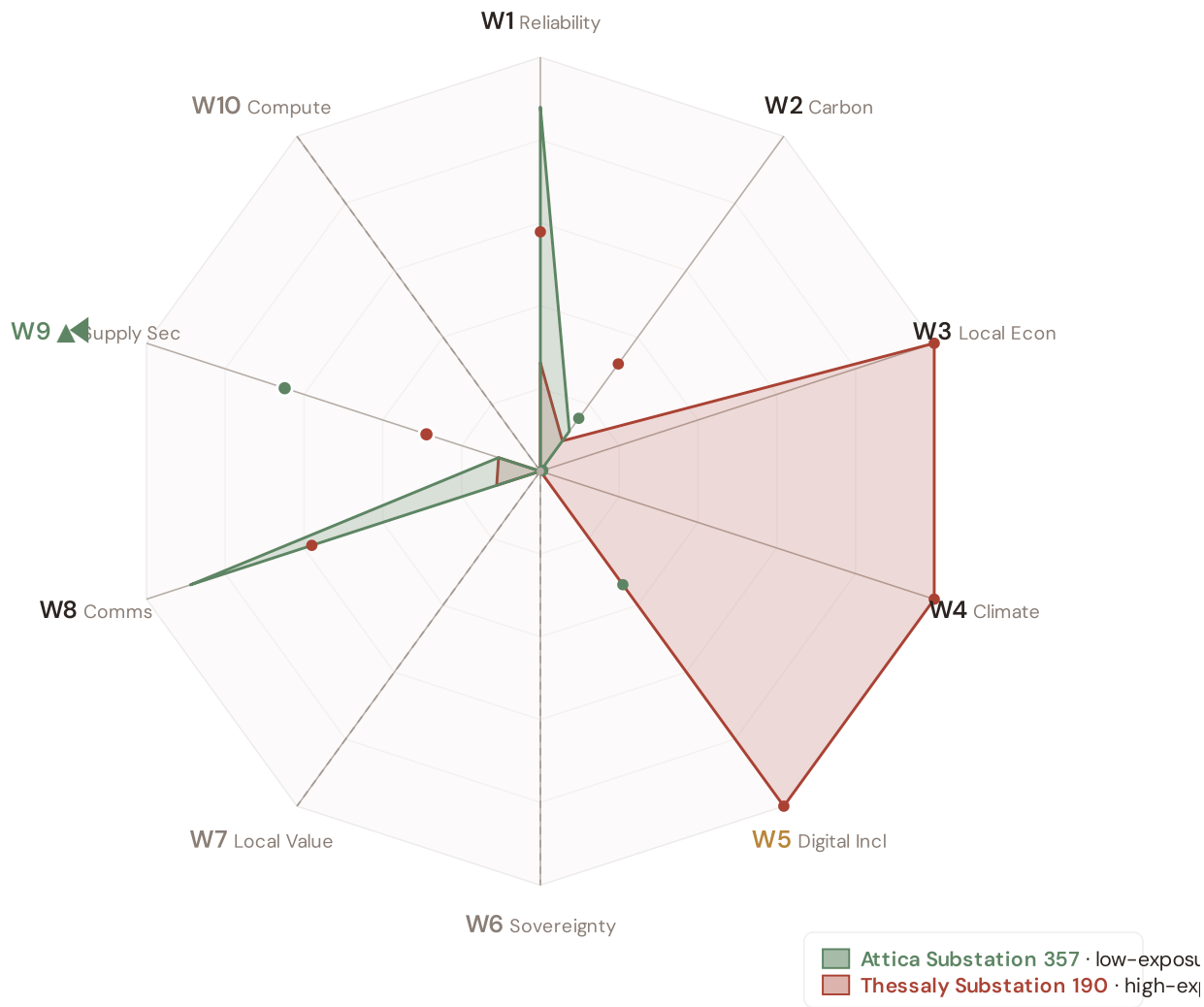
Per Convention #6 §5.6.7 academic-defensibility: the 4-tier classification aligns with IPCC AR6 + JRC COIN + Reckien (2021) continuum literature. The Published-baseline tier (W9) is the principled mesh resolution that honors both peer-review status + acknowledged scope limit without forcing a false binary.

v4.2 methodology disclosure — per-country normalisation

Across Greece, the substation fleet's component scores (C / V / I / E / S / T) and the Re composite have been normalised to [0, 1] via the canonical **P5/P95 percentile** method, using *per-country empirical bounds*. This is the same procedure Italy's v4.0.2 deployment applied at the scoring stage (SSI_v4_0 Italy/.../scoring-it/normalise.py); for v4.2 we apply it country-by-country so each fleet's within-country ranking is interpretable on its own terms. Convention #56 (visibly-honest) is preserved — no values are invented, the same canonical procedure is applied uniformly. Italy's pilot fleet keeps its native spread (0.17 – 0.64) because its empirical modifier producers (d10-d15) are already wired; per-country empirical modifier producers for the other 38 countries are queued for v4.5 and will refine the absolute risk ranking. Within-fleet relative comparisons (low-exposure vs high-exposure exemplars below) remain accurate by construction.

Radar 2 visualization — anti-maladaptation governance gate

Two worked examples overlaid: **Attica Substation 2** (low-exposure reference, Re_norm 0.000) and **Epirus Substation 5** (high-exposure reference, Re_norm 1.000). Axis labels are colour-coded per the 5/1/1/3 mesh resolution; the ▲ chevron marks W9 as Published-baseline. Hover any axis label for the methodology tooltip.



Attica Substation 357 polygon (sage): the low-exposure reference for this country — compact footprint reflects the fleet floor on the W-axis baselines. Thessaly Substation 190 polygon (terracotta): the high-exposure reference — larger footprint where this country's fleet's W-axis values reach the upper percentile. Commercial-tier axes (W6 data sovereignty, W7 local value retention, W10 compute multiplier) display at zero per Convention #6 §5.6 mesh design — no public-data computation. W2 carbon uses EU-average MEF fallback (0.36) per v4.2 calibration.

W1–W10 axis enumeration with audit-state classification

5/1/1/3 4-TIER · V4.2 MESH RESOLUTION · LIVE VALUES VIA
COMPUTE_W_BASELINES.PY

AXIS	NAME	AUDIT STATE	METHODOLOGY BASIS
W1	Reliability	Published	$C \times E2_{\text{local}} \times \text{population} (SAIDI + + \text{-POV})$
W2	Carbon	Published	$T1 \times \text{zonal MEF} \times (V \cdot \text{deg} / 100) (+ \text{EEA})$
W3	Local Economic	Published	$\text{norm}(R10_{\text{just}}) - \text{EU-SILC} + \text{OIPE energy-poverty}$
W4	Climate Resilience	Published	$(R6d - 1) + (R6e - 1) + (R6c - 1) + \frac{1}{2}(R9 - 1) - v4.2$ resilience modifier family
W5	Digital Inclusion	Baseline-only	$\text{norm}(R8_{\text{adapt}}) - \text{bulk SM} + + \text{DESI}$
W6	Sovereignty	Commercial	Foundation contact (ssi_index@ikenga.eu)
W7	Local Value Retention	Commercial	Foundation contact (ssi_index@ikenga.eu)
W8	Communications	Published	$C \times p_{\text{crit_10yr}} \times \text{population}$ (Markov 10-yr critical probability)
W9 ▲	Supply Security	Published-baseline ▲	$0.60 \times BC_{\text{pct}} + 0.30 \times \text{bridge} + 0.10 \times (1 - \text{deg} / 20)$ — OSM power-graph betweenness centrality + bridge identification + degree-based redundancy
W10	Compute Multiplier	Commercial	Foundation contact (ssi_index@ikenga.eu)

▲ W9 Published-baseline — mandatory transparency tooltip

W9 — Supply Security (Published-baseline). Public-tier methodology: betweenness centrality + bridge identification + degree-based redundancy, computed deterministically from OpenStreetMap power-graph data. Peer-reviewed academic basis: Albert et al. (2000) *Nature*; Holme et al. (2002); broad power-grid centrality literature. Acknowledged scope limit (per Buldyrev et al. (2010) *Nature* + IEEE 493): topology-only baseline does not capture N-1 contingency dynamics or interdependent-network cascades. Richer methodology — N-1 contingency simulation + flexibility-services valuation — lives in SSI-ENN L2-22 / L2-23 commercial deliverables. Mail to ssi_index@ikenga.eu for extended analysis.

Worked examples. The W1–W10 per-substation baseline values for two anchor cases — Attica Substation 357 (low-exposure reference) and Thessaly Substation 190 (high-exposure reference) — are computed live by `_audit-snapshot/scripts/d3_radar2_per_country.py` against the country's `ssi-data.json`. Per Convention #6

§2.3.6 Path B refactor (11 Jun 2026), this country’s exemplars are now algorithmically selected (min/max Re_norm) rather than named picks. See the FC v2 reference for W-axis methodology + commercial-tier rationale.

Anti-maladaptation discipline. The 10-axis Radar 2 baseline is the per-substation floor that any post-intervention scenario must protect. No W axis may degrade after a BESS deployment, line upgrade, or operational change. The discipline is enforced per axis (not in aggregate), and per substation (not in fleet average). This is the v4.2 governance gate that ensures grid-modernisation projects deliver net community value without hidden trade-offs.

SECTION G

Looking Ahead

NEXT MONTH — EDITION 05

Alpine Valleys

Deep-dive into Region 05’s grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Narrow valleys, long restoration MTTR.

NEXT DATA REFRESH

Q4 2026 — o Quarterly Sources

none are due for refresh in Q4. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **HEPI (Hellenic Energy Policy Index)** (3 variables → E2 energy transition metrics).

FLEET SPOTLIGHT

Seismic Zone 4–5: 255 Substations in Highest Hazard

255 substations sit in the highest seismic hazard zones (Zones 4–5, moderate to strong). Of these, **110** already score High/Critical — meaning their grid condition is poor *before* considering earthquake risk. The R6b seismic modifier captures this compound vulnerability.

Edition 05 will be published on the second Thursday of October 2026. Deep-dive region: **Region 05**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an Greek utility, , municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI’s credibility depends on open scrutiny.

Foundation contact

The SSI Index is a non-profit foundation project, please mail to ssi_index@ikenga.eu for more details.